



Particle adventure

An award winning interactive tour of quarks, gluons, matter, antimatter, neutrinos and extra dimensions. <https://digit.in/dec19-67>



The Planck epoch

For 10^{-35} seconds after creation, the laws of physics were still to come into play, in a particle desert <https://digit.in/dec19-68>

BARYON ASYMMETRY

Why is there more matter than antimatter?

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he universe came into existence 13.79 billion years ago, expanding from what is called a cosmic singularity.

Matter was crushed so densely together, that the temperature and the density of the universe at that point, can be considered to be infinite. The singularity from which the universe expanded was even more dense than the singularities at the hearts of black holes. Black holes are formed when the remnants of dense stars go through a

gravitational collapse, with the atomic structures collapsing into themselves. The singularity was expanding, and not of a fixed size as the remnant of a dead star is. This is why the universe did not experience a gravitational collapse as soon as it came into being.

In an unimaginably small fraction of a second after the singularity (10^{-37} seconds), the universe expanded and cooled, a period of time called the cosmic inflation. At this time, and even the subatomic particles had not come into existence. These are protons, neutrons and electrons. The substructure of these subatomic particles, which have never been observed in isolation, quarks and gluons were packed together in a dense soup called the quark gluon plasma. This state of the very early universe has been created in high energy particle accelerators such as the Large Hadron Collider (LHC) at CERN and Relativistic Heavy Ion Collider (RHIC) at Brookhaven

National Laboratory. The period in time is known as the quark epoch.

As the universe expanded and cooled further, the quarks clumped together to form the first hadrons, the most well known of which are the protons and neutrons. This is known as the hadron epoch. As the hadrons were being created, their corresponding anti-hadrons also came into existence, in pairs. For example, the antiparticle of the proton is the antiproton, and both would emerge in tandem. During the hadron epoch, any matter that came into contact with antimatter resulted in an annihilation, where both the particles were destroyed into high energy rays. Most of the matter that was created during the Big Bang, was destroyed within a second. The remnants of the annihilation later played a role in the lepton epoch that followed, approximately one second after the Big Bang. It was in the lepton epoch that particles such as electrons, neutrinos and muons were created. These were created in pairs too, the positively charged positron being an antiparticle to the negatively charged electron.